Online Anchor-Robust Forecasting under Economic Regime Shift: Immunising Streaming Predictors through an Online-Discovered Causal Channel

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Abstract

The prevailing way to forecast through economic regime shifts is *reactive*: monitor a statistic, declare that the world has changed, and recalibrate — paying a re-adaptation cost at every break and, by construction, always arriving a step late. This paper develops the opposite strategy. We argue that many economic regime shifts do not arrive out of nowhere but travel through an *observable causal channel* — a handful of regime drivers such as geopolitical risk, economic policy uncertainty and financial volatility — and that a forecaster can therefore be *immunised* against them in advance by removing the channel's influence on its errors, with no detection and no trigger. We turn this idea into a streaming algorithm, OARF (Online Anchor-Robust Forecasting), which at every step drives the component of its forecast residual that is correlated with the regime drivers toward zero; and, in its distinctive second layer, *discovers that channel online* from the data rather than requiring the analyst to name it. To evaluate such a method we introduce *interventional regret* — regret against the best regime-conditional predictor under the worst-case bounded future intervention on the regime driver — and prove that the fixed-channel learner is *no-regret* in this sense against a comparator that switches at most K times: the regret is $O(\sqrt{T(1+K)})$ with oracle anchor moments and $O((1+K)T^{2/3})$ for the fully online estimator, the latter being the price of learning the robustness geometry without changepoint detection. On a controlled structural-causal-model benchmark with a held-out grid of interventions $\mathrm{do}(A := \nu)$, OARF reduces the worst-case interventional error by 55–67% relative to reactive online learners and to the most recent (2025–2026) distributionally robust competitors, approaching a batch causal oracle while remaining fully online, and its discovery layer recovers the true intervention subspace with mean alignment 0.87. On sixteen years of daily energy-volatility data combined from a standard pool of heterogeneous forecasters, OARF is statistically indistinguishable from the best method by the Model Confidence Set yet consistently leads the 2026 robust combiners on the regime-robustness metrics, while exposing an interpretable, automatically discovered channel. The contribution is at once conceptual — robustness to the *cause* of a shift rather than reaction to its *symptom* — and practical: an anisotropic, causally shaped uncertainty set in place of the agnostic balls of distributionally robust optimisation.

Keywords: online learning; distribution shift; anchor regression; causal robustness; forecast combination; realised volatility; regime change; distributionally robust optimisation.

JEL: C53, C22, C58, Q47.

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1 Introduction

Economic and financial time series rarely keep still. They are punctuated by *regime shifts* — geopolitical shocks, monetary-policy turns, energy crises, and, in carbon markets, discrete regulatory interventions such as the activation of the EU Emissions Trading System Market Stability Reserve — each of which moves the relationship a forecaster is trying to track. A model that was well calibrated yesterday can be badly miscalibrated tomorrow, and the dominant response in the online-learning and conformal-prediction literatures is to treat this as a detection problem: watch a loss or a coverage statistic, raise an alarm when it drifts, and then recalibrate or re-weight [8, 11, 14]. This reactive posture is intuitive and often effective, but it has two structural weaknesses. It is always one step behind — the alarm can only fire after the loss has already risen — so it pays a re-adaptation cost at every break; and it treats the *symptom* of a shift, a deterioration in fit, as if it were the *cause*.

We take a different view. In many economic settings the cause of a shift is not hidden: it is an *observable channel* of regime drivers whose own changing distribution is precisely what perturbs the forecasting environment. When geopolitical risk spikes or policy uncertainty jumps, the joint distribution of prices, volatilities and their predictors moves with it. If a forecaster's residual can be made *invariant* to that channel — if the part of its error that co-moves with the drivers is held at zero — then the forecaster is protected against shifts in the channel's distribution *before* they happen, without any detection step. This is exactly the logic of anchor regression [15], which trades a little in-sample fit for invariance to interventions on a designated "anchor" variable and thereby minimises a worst-case risk over bounded interventions. Anchor regression, however, is a batch estimator that assumes the analyst already knows which variables constitute the anchor. Neither assumption fits a streaming economic deployment, where data arrive one observation at a time and where the right channel — which of the many candidate drivers actually carries the regime — is itself uncertain.

This paper closes both gaps. We develop OARF, a fully streaming anchor-robust forecaster whose first-order update augments the ordinary prediction gradient with an anchor-robustness term assembled from exponential-moving-average cross-moments; with the robustness weight set to zero it degenerates exactly to online gradient descent, so robustness is a single, interpretable dial. On top of it we build a second layer that *learns the channel from the stream*, identifying the subspace of the observable context along which the relationship to the regime is most unstable, and feeding that learned anchor back into the robust update on a slower timescale. Finally, because the usual notions of regret do not capture what such a method is for, we define and analyse *interventional regret*: regret measured against the best regime-conditional predictor under the worst-case bounded future intervention.

Contributions.

1. **A streaming anchor-robust learner** (Section 3). OARF performs causal-robust forecasting online through a first-order update whose robustness term is built from EMA cross-moments of *centred* variables; centring removes the regime mean-shift — the intervention itself — and leaves the within-regime covariance that shapes the robustness set. Setting the penalty $\xi = 0$ recovers online gradient descent exactly.
2. **Online intervention-channel discovery** (Section 3). A two-timescale rule learns *which directions* of the observable context act as the anchor, by maximising the across-regime dispersion of the context subject to a norm constraint — lifting the method above "a known batch estimator run online" and making it usable when the analyst cannot name the channel.

3. **Interventional regret and a guarantee** (Section 3). We formalise regret against the best regime-conditional predictor under the worst-case bounded intervention $\mathrm{do}(A := \nu)$ and prove, with a complete argument, that the fixed-channel learner achieves sublinear interventional regret against a K-switch comparator with no prior knowledge of K — $O(\sqrt{T(1+K)})$ with oracle anchor moments, $O((1+K)T^{2/3})$ for the online estimator — and we are explicit about why the comparator must be regime-conditional rather than adversarial [17].

4. **A complete, reproducible evaluation** (Sections 4–5). We pair a controlled structural-causal-model benchmark, in which a held-out grid of interventions makes interventional robustness measurable against ground truth, with sixteen years of public daily data; we compare against eleven baselines spanning classical floors, the 2025–2026 state of the art in distribution-shift adaptation and distributionally robust optimisation, and causal-robust ablations; and we report a full battery of regime-aware, interventional, distributional and decision-economic metrics with Diebold–Mariano [5] and Model Confidence Set [9] significance, alongside a complete ablation of every design choice.

The empirical message is sharp and, we think, intuitive. When regime shifts genuinely travel through an observable channel, immunisation buys a large reduction in worst-case interventional error for only a small in-distribution price — a clean adaptation–immunisation trade-off that the reactive and agnostic-robust methods, seeing the same data, simply cannot reach, because they hedge in the wrong geometry or only after the fact.

2 Related work

Our method sits at the meeting point of three literatures, and it is easiest to position by what it borrows from and where it departs from each.

Causal robustness and invariance. The intellectual root is the observation that a predictor which is *invariant* across environments is also robust to interventions that move between them. Invariant causal prediction [13] makes this precise for the identification of causal parents, and anchor regression [15] turns it into an estimator: by penalising the projection of the residual onto an anchor variable it interpolates between ordinary least squares and an instrumental-variables solution, and the penalised problem is provably equivalent to minimising worst-case risk over a bounded set of interventions on that anchor. DRIG [16] generalises the principle to distributional robustness through invariant gradients. What all of these share — and what we must overcome for a streaming deployment — is that they are batch procedures that take the environment or anchor variable as given. OARF keeps the worst-case-intervention semantics of anchor regression but makes the estimator online and, crucially, learns the anchor channel from data.

Reactive online adaptation. A second, very active line treats distribution shift as something to detect and then accommodate. Adaptive conformal inference [8] adjusts a miscoverage level online to maintain coverage under drift. The 2025 state of the art sharpens the detect-then-adapt loop considerably: M-FISHER [11] maintains an exponential test martingale over non-conformity scores and, when the martingale crosses a Ville threshold, takes a Fisher-preconditioned natural-gradient correction; WATCH/WCTM [14] monitors a *weighted* conformal test martingale that can tolerate benign covariate shift while still flagging harmful concept shift, and adapts the prediction interval accordingly. These methods are the natural incumbents against which a proactive method must be judged — and the comparison is precisely the point, since OARF aims to make the trigger unnecessary by removing the channel's influence before it can raise the loss.

Distributionally robust forecasting (2026). The closest competitors are recent methods that, like OARF, optimise an explicitly worst-case objective. Wang [18] propose online randomised distributionally robust forecast combination for dependent data, drawing combination weights from a distribution over the simplex and updating it by stochastic mirror descent on a Wasserstein ambiguity objective. Liu and Cheng [12] develop adaptive robust combination with variance- and Expected-Shortfall-scaled exponential weights for yield-curve forecasting, and Chen et al. [3] analyse Wasserstein distributionally robust *online* learning as a primal–dual game with regret guarantees; a parallel strand makes conformal decisions cost-sensitive [19]. The decisive distinction is geometric. These methods hedge over an *isotropic* ambiguity ball centred at the empirical distribution: they spread robustness equally in all directions, including directions along which the world never actually moves. OARF instead hedges over the *anisotropic ellipsoid carved out by the causal channel*, $\mathbb{E}[aa^\top]$, spending its robustness budget only where shifts genuinely travel. Our experiments are designed to isolate exactly this difference, and they show it to be the source of OARF's advantage.

Online convex optimisation. Methodologically, the regret analysis rests on dynamic-regret results for online gradient descent [10, 21], and the implementation uses AdaGrad-style adaptive preconditioning [6] so that a single step size is scale-invariant across heterogeneous features. The economic measurements draw on the daily Geopolitical Risk index [2] and Economic Policy Uncertainty index [1], on realised-volatility modelling in the heterogeneous-autoregressive tradition [4], and on the volatility-timing logic of Fleming et al. [7] for the decision-economic evaluation.

3 The OARF method and its guarantee

Setup and a leak-free protocol. We work in discrete time $t = 1, \ldots, T$. At each step the forecaster observes predictors $x_t \in \mathbb{R}^d$ — which in a pure forecasting problem are explanatory variables and in a combination problem are the forecasts of several base models — together with a vector of candidate regime drivers $z_t \in \mathbb{R}^p$, and it must predict a scalar target y_t. A linear predictor with feature map $\phi(x) = [1; x] \in \mathbb{R}^{d+1}$ and weights w emits $\hat{y}_t = w^\top \phi(x_t)$, incurs the squared loss $\ell_t(w) = (y_t - \hat{y}_t)^2$, and writes its residual $r_t = y_t - \hat{y}_t$. The *anchor* is a (possibly learned) linear read-out of the context, $a_t = B^\top z_t \in \mathbb{R}^q$, with channel matrix $B \in \mathbb{R}^{p \times q}$. The entire pipeline is leak-free: at step t we first predict from x_t, then reveal y_t and update, and every standardisation uses past-only statistics; when we later evaluate a frozen learner on held-out interventions we replay exactly that frozen, past-only standardiser, so no future information can ever enter a forecast.

The causal-robustness objective. Following anchor regression [15], OARF trades raw predictability against invariance of the residual to the anchor by minimising

$$b_{\mathrm{AR}}(w;\xi) = \underbrace{\mathbb{E}\big[(y - w^\top\phi(x))^2\big]}_{\text{predictive loss}} + \xi\, \underbrace{\mathbb{E}[ar]^\top \big(\mathbb{E}[aa^\top]\big)^{-1}\mathbb{E}[ar]}_{P(w)=\|\operatorname{proj}_a r\|^2}, \qquad \xi \geq 0. \tag{1}$$

The penalty $P(w)$ is the squared length of the projection of the residual onto the anchor; driving it down forces $\mathbb{E}[ar] \to 0$, so the predictor stops exploiting any component of the signal that co-moves with the regime drivers. At $\xi = 0$ the objective is ordinary least squares, and as ξ grows the solution moves toward the invariant predictor. The reason this is the right object is not merely intuitive:

Proposition 1 (Interventional-robustness equivalence, 15). *For linear structural causal models there exists a radius $\rho(\xi)$, increasing in ξ, such that*

$$\min_{w} b_{\mathrm{AR}}(w;\xi) \;=\; \min_{w} \;\sup_{\nu:\, \|\nu\|\le\rho(\xi)}\; \mathbb{E}_{\,\mathrm{do}(a:=\nu)}\big[(y - w^\top\phi(x))^2\big]. \tag{2}$$

In words, minimising the ξ-penalised loss is identically minimising the worst-case mean-squared error over all bounded interventions $\mathrm{do}(a := \nu)$ on the anchor. The set being hedged over is an ellipsoid shaped by the anchor covariance $\mathbb{E}[aa^\top]$ — not the isotropic ball of the distributionally robust methods of Section 2 — and this is the geometric distinction our evaluation sets out to test.

A streaming anchor-gradient update. The gradient of (1) in the coefficient block separates into a prediction term and a robustness term,

$$\nabla_w b_{\mathrm{AR}} = -2\,\mathbb{E}[\phi r] \;-\; 2\xi\,\mathbb{E}[\phi a^\top]\big(\mathbb{E}[aa^\top]\big)^{-1}\mathbb{E}[ar], \tag{3}$$

and the move from batch to streaming is to replace each population moment by an exponential moving average (decay β) of its sample analogue, computed on *centred* variables. Centring is not a cosmetic detail: subtracting the running means removes the across-regime mean-shift in the anchor — the very thing an intervention does — and leaves the within-regime covariance, which is what the robustness term (2) is supposed to use. Writing $\tilde{\cdot}$ for the EMA-centred quantities, we maintain running estimates m_t^{Xr}, m_t^{XA}, M_t^{AA} and m_t^{Ar} of $\mathbb{E}[\tilde{x}\tilde{r}]$, $\mathbb{E}[\tilde{x}\tilde{a}^\top]$, $\mathbb{E}[\tilde{a}\tilde{a}^\top]$ and $\mathbb{E}[\tilde{a}\tilde{r}]$, and update

$$\boxed{w_{t+1} = w_t - \eta\Big[\underbrace{-r_t\phi(x_t)}_{\text{prediction}}\underbrace{-\,2\xi\,[\,0;\, m_t^{XA}(M_t^{AA}+\epsilon I)^{-1}m_t^{Ar}\,]}_{\text{anchor robustness}}+\lambda_2 w_t\Big],} \tag{4}$$

with the intercept left unpenalised, a ridge ϵ stabilising the anchor Gram and an ℓ_2 term λ_2. In practice we take the step with AdaGrad-style per-coordinate scaling [6]: the *gradient* is exactly (4), and only the step length is preconditioned, which leaves the online-convex-optimisation regret machinery intact while making one global learning rate behave sensibly across features of very different scale. The decay β tunes how quickly the moment estimates forget, and so how tightly they track the current regime.

Discovering the channel online. Everything so far assumes the analyst has supplied the channel B. The layer that makes OARF more than "anchor regression, online" removes that assumption. The guiding observation is that, under the anchor model, a regime acts on the observable context by displacing the anchor's location; the directions that carry the regime are therefore exactly the directions of z whose conditional mean is *most unstable across regimes*, while genuinely exogenous or nuisance directions stay put. We therefore learn B, on a slow timescale $\eta_B \ll \eta$, by projected gradient ascent on the across-regime dispersion of the context's conditional mean,

$$\max_{\|B\|_F\le 1}\; \mathcal{D}(B) = \operatorname*{Var}_{\text{regimes}}\big(\mathbb{E}[B^\top z \mid \text{regime}]\big) - \gamma_c\,\big\|\mathbb{E}[B^\top z]\big\|^2, \tag{5}$$

where the second term penalises loading on the stable, grand-mean component and so keeps the recovered channel exogenous-like. Streaming this objective requires a proxy for "regime": we use a forgetting partition of the stream into length-L blocks, long enough that the short-horizon autocorrelation of a stationary nuisance driver averages out within a block but short enough that genuine regime changes fall across block boundaries. Letting S be the across-block scatter of the block-mean context and u its grand mean, the ascent direction is $\nabla_B\mathcal{D} = 2(S - \gamma_c uu^\top)B$, after which B is re-orthonormalised onto the Frobenius ball; a lightweight non-robust reference predictor is carried alongside to keep discovery oriented toward predictively relevant directions. Algorithm 1 states the complete loop, with the discovery layer as an optional inner block.

Algorithm 1 OARF with online channel discovery (one streaming pass)

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input: penalty $\xi$, rates $\eta, \eta_B$, decay $\beta$, ridge $\epsilon, \lambda_2$, block $L$, channel $B$ (or random init if learning)
$w \leftarrow 0$;  EMA means and cross-moments $\leftarrow 0$;  block buffer $\leftarrow 0$
for $t = 1, \dots, T$ do
    observe $x_t, z_t$;  standardise (past-only, winsorised): $x_s, z_s$;  $a \leftarrow B^\top z_s$
    $\hat{y}_t \leftarrow w^\top [1; x_s]$;  emit $\hat{y}_t$;  reveal $y_t$;  $r \leftarrow y_t - \hat{y}_t$
    update EMA means; centre $\tilde{x}, \tilde{a}, \tilde{r}$; update $m^{Xr}, m^{XA}, M^{AA}, m^{Ar}$
    $g \leftarrow -r[1; x_s]$;   $g_{1:} -= 2\xi\, m^{XA}(M^{AA} + \epsilon I)^{-1} m^{Ar} - \lambda_2 w_{1:}$
    $w \leftarrow w - \text{AdaGradStep}(g)$
    if learning the channel then
        accumulate $z_s$ into the block buffer; every $L$ steps update across-block scatter $S$, mean
        $u$, and set $B \leftarrow \text{orth}\big(B + \eta_B\, 2(S - \gamma_c u u^\top)B\big)$
    end if
    absorb $(x_t, z_t)$ into the standardiser statistics
end for
```

Interventional regret. To state what OARF guarantees we need a regret notion that rewards robustness to *future* interventions, not merely good tracking of the past. For a radius $\rho \geq 0$ define

$$\mathrm{IntReg}_T(\rho) = \sum_{t=1}^{T} \ell_t(w_t) - \min_{u \in \mathcal{W}} \sum_{t=1}^{T} \sup_{\|\nu_t\| \leq \rho} \mathbb{E}_{\mathrm{do}(a:=\nu_t)}\big[(y_t - u_t^\top \phi(x_t))^2\big] : \tag{6}$$

the learner's realised loss, measured against the best comparator in a class $\mathcal{W}$ that is itself stress-tested by the worst bounded intervention at every step. We make four assumptions, each mild and each used in a specific place.

Assumption 1. *(Boundedness.) The standardised, winsorised features and anchor are bounded, so the per-step gradients of the surrogate below are bounded in norm by a constant G, and the comparator class $\mathcal{W}$ lies in a Euclidean ball of radius D.*

Assumption 2. *(Linear-SCM anchor model.) The data follow a linear structural causal model in which the anchor enters as in Section 3, so the equivalence of Proposition 1 holds within each regime, and the radius is adequate, $\rho(\xi) \geq \max_t \|\nu_t^\star\|$, where $\nu_t^\star$ is the realised regime's anchor displacement (so each realised regime is one admissible intervention in the ball).*

Assumption 3. *(Oblivious regime schedule and comparator.) The regime schedule is exogenous and oblivious to the learner; the per-step worst-case risk ϕ_t defined below is therefore a deterministic function of w, and $\mathcal{W}$ is the class of regime-conditional comparators that switch at most K times (path length $P_T \leq 2DK$).*

Assumption 4. *(Within-regime stationarity.) Within each regime the joint distribution of (ϕ, a, y) is stationary with mixing fast relative to the EMA window, so the EMA moment estimates obey the concentration of Lemma 3.*

The proof rests on viewing Algorithm 1 as online gradient descent, fed *inexact* gradients, on a sequence of convex per-step risks; we isolate the three ingredients as lemmas and then assemble them. Throughout, $\phi_t(w) := \sup_{\|\nu\| \leq \rho(\xi)} \mathbb{E}_{\mathrm{do}(a:=\nu)}[(y_t - w^\top \phi(x_t))^2]$ is the per-step worst-case risk appearing in (6).

Lemma 1 (Convex surrogate and gradient decomposition)**.** *ϕ_t is convex, and under Assumption 2 it equals, within the active regime, the anchor-penalised risk $\phi_t(w) = \mathbb{E}_t[(y - w^\top \phi)^2] + \xi P_t(w)$ of (1). The step direction $\hat{g}_t$ taken by Algorithm 1 satisfies $\hat{g}_t = \nabla\phi_t(w_t) + \zeta_t$, where the error decomposes as $\zeta_t = M_t + b_t$ with $M_t := -r_t\phi(x_t) - \mathbb{E}_t[-r\phi]$ a martingale difference ($\mathbb{E}[M_t \mid \mathcal{F}_{t-1}] = 0$) and b_t the deterministic ($\mathcal{F}_{t-1}$-measurable) error of the EMA-estimated penalty gradient relative to $\nabla(\xi P_t)$.*

Proof. For each fixed ν, $\mathbb{E}_{\mathrm{do}(a:=\nu)}[(y - w^\top\phi)^2]$ is a convex quadratic in w; a pointwise supremum of convex functions is convex, giving convexity of ϕ_t. The stated identity is Proposition 1 applied within the (stationary) active regime. For the gradient, the prediction term $-r_t\phi(x_t)$ is an unbiased one-sample estimate of $\frac{1}{2}\nabla\mathbb{E}_t[(y - w^\top\phi)^2] = -\mathbb{E}_t[r\phi]$, whose centred part is the martingale difference M_t; the robustness term of (4) substitutes the EMA moments $m_t^{XA}, (M_t^{AA} + \epsilon I)^{-1}, m_t^{Ar}$ — all $\mathcal{F}_{t-1}$-measurable — for the population moments in $\nabla(\xi P_t)$, and b_t is exactly that substitution error. □

Lemma 2 (Inexact dynamic regret)**.** *Run projected online gradient descent with a path-length-adaptive step-size wrapper (Ader; 20) on the convex losses ϕ_t, but fed the inexact gradients $\hat{g}_t$. Then for any comparator sequence $u_{1:T} \subset \mathcal{W}$ with path length P_T,*

$$\sum_{t=1}^{T} \phi_t(w_t) - \sum_{t=1}^{T} \phi_t(u_t) \;\le\; O\big(\sqrt{T(1+P_T)}\big) \;+\; \sum_{t=1}^{T} \langle \zeta_t,\, w_t - u_t \rangle,$$

where the first term is attained without *prior knowledge of P_T (hence of K).*

Proof. With exact gradients, Ader attains the minimax dynamic regret $O(\sqrt{T(1+P_T)})$ for convex, G-Lipschitz losses over a radius-D set, adaptively in P_T [20]; this already subsumes the single-step bound of Hazan [10], Zinkevich [21]. Replacing the exact gradient by $\hat{g}_t = \nabla\phi_t(w_t) + \zeta_t$ adds, through the standard first-order regret telescoping, exactly the linear-in-error term $\sum_t \langle \zeta_t, w_t - u_t\rangle$ (convexity gives $\phi_t(w_t) - \phi_t(u_t) \le \langle \nabla\phi_t(w_t), w_t - u_t\rangle = \langle \hat{g}_t, w_t - u_t\rangle - \langle \zeta_t, w_t - u_t\rangle$, and the $\hat{g}_t$ terms are bounded by the exact-gradient analysis). □

Lemma 3 (Gradient-error control)**.** *Under Assumptions 1–4, and against the (deterministic, by Assumption 3) comparator that minimises (6): (i) $\mathbb{E}\big[\sum_t \langle M_t, w_t - u_t\rangle\big] = 0$; and (ii) the EMA penalty-gradient error obeys $\mathbb{E}\|b_t\| \le c_1\sqrt{1-\beta}$ at steps that lie at least one window $1/(1-\beta)$ into a regime, while within $1/(1-\beta)$ steps of each of the K regime changes $\|b_t\| \le c_2$; consequently $\sum_t \mathbb{E}\|b_t\| \le c_1 T\sqrt{1-\beta} + c_2 K/(1-\beta)$. With* oracle *moments, $b_t \equiv 0$.*

Proof. (i) w_t is $\mathcal{F}_{t-1}$-measurable and, since the regime schedule is oblivious (Assumption 3), the minimising comparator u_t is a deterministic sequence; hence $\mathbb{E}[\langle M_t, w_t - u_t\rangle \mid \mathcal{F}_{t-1}] = \langle \mathbb{E}[M_t \mid \mathcal{F}_{t-1}], w_t - u_t\rangle = 0$, and the claim follows by the tower rule. (ii) An EMA with decay β of a stationary sequence has variance $\Theta(1-\beta)$ about its mean and bias decaying geometrically in the elapsed within-regime horizon; by Assumption 1 the penalty gradient is a Lipschitz function of these moments (the ϵ-ridge bounds the inverse), so its estimation error inherits the $\Theta(\sqrt{1-\beta})$ steady-state scale. Immediately after a regime change the EMA still mixes the previous regime for $O(1/(1-\beta))$ steps, over which the bounded gradients give $\|b_t\| \le c_2$. Summing the two regimes of behaviour over the stream gives the stated bound; oracle moments make the substitution error vanish identically. □

Theorem 1 (Fixed-channel interventional regret)**.** *Under Assumptions 1–4, Algorithm 1 with a fixed channel and the Ader step wrapper attains* sublinear *expected interventional regret against any K-switch comparator, with no prior knowledge of K:*

(i) with oracle anchor second moments, $\mathbb{E}\big[\mathrm{IntReg}_T(\rho(\xi))\big] = O\big(\sqrt{T(1+K)}\big)$*;*

(ii) with the EMA moment estimates of Algorithm 1 and decay $1-\beta_T = T^{-2/3}$*,* $\mathbb{E}\big[\mathrm{IntReg}_T(\rho(\xi))\big] = O\big((1+K)\,T^{2/3}\big)$*.*

Proof. Combining Lemmas 1–3 and taking expectations, for the deterministic minimising comparator $u_{1:T} \in \mathcal{W}$ (path length $P_T \leq 2DK$),

$$\mathbb{E}\Big[\textstyle\sum_t \phi_t(w_t)\Big] - \textstyle\sum_t \phi_t(u_t) \;\leq\; O\big(\sqrt{T(1+K)}\big) + \underbrace{\mathbb{E}\big[\textstyle\sum_t \langle M_t, w_t - u_t\rangle\big]}_{=0} + D\textstyle\sum_t \mathbb{E}\|b_t\|.$$

With oracle moments the last term is zero, giving (i). With EMA moments, $D\sum_t \mathbb{E}\|b_t\| \leq D\big(c_1 T\sqrt{1-\beta} + c_2 K/(1-\beta)\big)$; setting $1-\beta_T = T^{-2/3}$ makes this $O(T^{2/3}) + O(KT^{2/3}) = O((1+K)T^{2/3})$, which dominates the $\sqrt{T(1+K)}$ term, giving (ii). Finally, by Assumption 2 the realised regime is an admissible intervention, so $\mathbb{E}_t[\ell_t(w_t) \mid \mathcal{F}_{t-1}] \leq \phi_t(w_t)$ and hence $\mathbb{E}[\sum_t \ell_t(w_t)] \leq \mathbb{E}[\sum_t \phi_t(w_t)]$; since $\min_{u\in\mathcal{W}} \sum_t \phi_t(u)$ is the comparator in (6), the two displays give $\mathbb{E}[\mathrm{IntReg}_T(\rho(\xi))]$ no larger than the stated rates. Both are $o(T)$, so the per-round regret vanishes. □

Remark 1 (Reading the two rates)*.* The gap between (i) and (ii) is the price of *learning the robustness geometry online without changepoint detection.* A detection-and-reset estimator could drive the moment error down at the $O(1/\sqrt{n})$ within-regime rate and recover (i), but only by reintroducing exactly the triggers OARF is designed to avoid; the $T^{2/3}$ rate is what a purely proactive, exponentially-forgetting estimator pays, and it remains sublinear. The deployed implementation uses a single AdaGrad instance rather than the Ader wrapper, which removes only the K-adaptivity and is what one wants in practice.

Remark 2 (What the guarantee does and does not claim)*.* Two qualifications matter. First, risk-adjusted no-regret is attainable only under stochastic, stationary or ergodic data, not against a fully adversarial environment [17]; this is why $\mathcal{W}$ is regime-conditional rather than adversarial and why observable regime drivers must enter the statement — they define the regimes the comparator may condition on. Second, anchor-based methods deliver *interventional robustness, not causal identification*: the estimand is the diluted-causal parameter, and we do not claim to recover a structural coefficient.

4 Experimental design

We evaluate OARF in two complementary arenas. A controlled structural-causal-model benchmark lets us measure interventional robustness against ground truth, because we can intervene on the anchor by construction; a sixteen-year real-data study then asks whether the mechanism survives contact with genuine, messy economic series, where no ground-truth intervention exists and the regime structure is milder. The two together are designed so that the synthetic study isolates *why* the method should help and the real study tests *whether* it does.

A controlled anchor-SCM with a measurable intervention gap. The synthetic generator (Table 1) is built so that the difference between ordinary prediction and interventional robustness is not a matter of taste but a quantity we can read off. An exogenous anchor A, whose mean shifts across eight regimes — those shifts *are* the in-sample interventions — and a hidden confounder H drive a block of causal *parents* X_{par} of the target, which is generated as $Y = X_{\mathrm{par}}^\top b_{\mathrm{par}} + H^\top d_H + \varepsilon_Y$. A second block of *children* is then generated downstream of the target, $X_{\mathrm{ch}} = Y c_{\mathrm{ch}} + AW_{Ac} + HW_{Hc} + \varepsilon_{\mathrm{ch}}$, so that the children are anti-causal predictors that also load on the anchor. The design has a deliberate trap. In-sample the children are extremely informative about Y — they are generated from it — so ordinary least squares and online gradient descent lean on them heavily; but a child's relationship to Y is contaminated by the anchor term AW_{Ac}, and under an intervention $\mathrm{do}(A := \nu)$ that term is set externally, so the child turns from informative to misleading and injects an error that grows with $\|\nu\|$. The only predictor that survives the intervention is the one that uses the parents alone, and that is precisely the gap anchor robustness is meant to close. To give the discovery layer something to recover, the observable context z is a random rotation of the anchor together with $p - q$ stationary AR(1) decoy drivers, so that there is a ground-truth channel B_{true} with $ZB_{\mathrm{true}} = A$. We hold out a frozen grid of interventions — ten magnitudes, six directions, four hundred draws each — spanning anchor values *beyond* the training range, and it is on this grid that every interventional number in Section 5 is computed.

Table 1: Configuration of the synthetic anchor-SCM (defaults; results average over 8 random seeds).

Symbol	Meaning	Value
T	stream length	8000
$d_{\mathrm{par}}, d_{\mathrm{ch}}$	causal parents / anti-causal children	$3, 3$
p	observable context dimension	8
q	true anchor (channel) dimension	2
h	hidden-confounder dimension	2
#regimes	anchor-mean regimes ($\Rightarrow$ 7 changepoints)	8
σ_{shift}	across-regime anchor-mean scale	1.5
σ_A	within-regime anchor sd	0.6
c_{ch} scale	strength of the target$\rightarrow$child edge	1.3
W_{Ac} scale	anchor$\rightarrow$child contamination	1.1
confounding	strength of the $H \rightarrow (X, Y)$ path	0.9
σ_X, σ_Y	predictor / target noise	$0.4, 0.5$

The real panel and the forecasting task. For the real study we assemble a public daily panel spanning 2010–2026 (Table 2), aligned to a business-day calendar with capped forward-fill. The targets are the two energy commodities most exposed to the geopolitical and policy regimes we care about, Brent crude and Henry Hub natural gas, and the prediction problem is one-step-ahead *log realised variance*. We choose volatility rather than returns deliberately: daily returns are very close to unforecastable, so a return study would mostly measure noise, whereas realised volatility is both genuinely predictable, through volatility clustering, and strongly regime-dependent, which is exactly the territory where regime-robustness can matter. The macro drivers — equity-market volatility (VIX), the dollar, the ten-year yield, and the daily Geopolitical Risk and Economic Policy Uncertainty indices — serve as the observable context, and the levels of the three uncertainty indices form the hand-chosen anchor for the fixed-channel variant, while OARF-CD is left to find its own. Regimes for the regime-aware metrics are defined, without look-ahead, as terciles of a past-only volatility-stress index, giving calm, normal and turbulent states and the transitions between them.

The forecast-combination pool. In the real study every method combines the same pool of one-step-ahead volatility forecasts, and the pool is chosen to be the standard one rather than anything bespoke. It comprises the heterogeneous-autoregressive cascade of daily, weekly and monthly realised variance [4]; an exponentially weighted RiskMetrics estimator; a GARCH$(1,1)$ conditional variance; the option-implied VIX as a forward-looking proxy; and a regularised regression on these together with the macro drivers. These are not interchangeable: HAR encodes long-memory persistence, the EWMA and GARCH terms capture short-run clustering with different decay structures,

Table 2: The public real-data panel. All series are free and citation-requested and are fetched by the released download script.

Series	Source / ticker	Span (daily)	Role
Brent crude	FRED `DCOILBRENTEU`	2010–2026	target (log-RV)
Henry Hub gas	FRED `DHHNGSP`	2010–2026	target (log-RV)
VIX	FRED `VIXCLS`	2010–2026	driver (level + change)
EUR/USD	FRED `DEXUSEU`	2010–2026	driver
10y Treasury	FRED `DGS10`	2010–2026	driver
Geopolitical Risk	Caldara–Iacoviello [2]	2010–2026	driver / anchor
Policy Uncertainty	Baker–Bloom–Davis [1]	2010–2026	driver / anchor
EUA carbon	manual export (optional)	—	optional target

and the implied-vol proxy injects forward-looking information that the backward-looking models miss. No single one dominates across regimes, which is precisely why combining them is worthwhile and why this is the natural arena in which the 2026 distributionally robust combiners were proposed. Every member of the pool is strictly causal: each forecast for day t is computed from information available through $t-1$.

Baselines. We compare against eleven methods, grouped not by superficial type but by the question each is meant to answer about OARF (Table 3). The *floors* — online gradient descent, which is literally OARF with $\xi = 0$; a rolling-window ridge regression; and adaptive conformal inference [8] — pin down what accuracy looks like with no robustness mechanism and what a strong batch refit can do, and so calibrate both ends of the scale. The *2025 reactive* methods, M-FISHER [11] and WATCH/WCTM [14], embody the detect-then-adapt paradigm that OARF is conceived to replace; placing them side by side tests directly whether removing a channel's influence in advance beats reacting to it after the loss rises. The *2026 distributionally robust* methods — randomised Wasserstein combination [18], the variance- and ES-scaled exponential-weight combiners [12], Wasserstein online learning [3] and cost-aware conformal inference [19] — are the most demanding comparison, because they too optimise a worst-case objective; what separates them from OARF is only the *shape* of the set they hedge over, an agnostic ball rather than a causal ellipsoid, so the comparison isolates the value of the causal geometry. Finally, the *batch causal-robust* estimators, anchor regression [15] and DRIG [16], use the same invariance principle as OARF but offline and with the channel handed to them; they therefore serve as an oracle upper bound on what the online, channel-learning version can hope to match. Every method is run under the identical leak-free protocol, with AdaGrad-stabilised steps where applicable.

Metrics and protocol. We report point accuracy (overall and root MSE, and the regime-aware refinements of per-regime, worst-regime and post-changepoint MSE over 50-step windows); the interventional robustness that is the paper's headline, namely the robustness curve MSE(ν) as a function of intervention magnitude and the worst-case error over the held-out grid; distributional quality for the methods with a quantile head (coverage, CRPS and interval width); and the economic value of a volatility forecast through a volatility-timing strategy [7]. Statistical significance is assessed pairwise with the Diebold–Mariano test [5] and jointly with the Model Confidence Set [9]. Throughout we use an expanding-window protocol in which the first fifth of each series is reserved for warm-up and hyper-parameter selection and the remainder is evaluated; synthetic numbers average over eight seeds and real numbers over rolling origins, and no future information

Table 3: The eleven comparison methods, organised by the question each answers about OARF. The column "Hedges over" names the uncertainty geometry; OARF is the only method whose set is an anisotropic *causal* ellipsoid, and the only proactive (non-triggered) robust learner.

Method	Year	Family	Hedges over	Posture
OARF/OARF-CD	ours	anchor-robust OCO	causal ellipsoid $\mathbb{E}[aa^\top]$	proactive
OGD	2003	floor	—	—
Rolling-OLS	—	floor	—	windowed
ACI	2021	conformal	coverage level	reactive
M-FISHER	2025	test-time adaptation	— (martingale trigger)	reactive
WATCH/WCTM	2025	conformal martingale	interval width	reactive
DR-Combo	2026	robust combination	Wasserstein ball	mirror descent
FC-DRO(-ES)	2026	robust combination	variance / ES	exponential weights
W-DRO-OL	2026	robust online learning	Wasserstein ball	primal–dual
CostACI	2026	cost-aware conformal	coverage level	reactive
AnchorReg / DRIG	'21/'23	causal (batch oracle)	anchor / environments	windowed

enters any forecast.

5 Results

Synthetic: immunisation against interventions. The controlled benchmark (Table 4 and Figures 1–4) delivers the paper's thesis in its clearest form, and the most informative way to read it is to notice that the in-distribution and the interventional rankings are almost mirror images. *In distribution*, where every regime has been seen, the batch and reactive methods are most accurate: rolling-window OLS, DRIG and the conformal heads post the lowest overall MSE, and OARF, which deliberately declines to use the anchor-correlated child signal, pays a modest premium for doing so. *Under intervention*, the order reverses. OARF attains a worst-case interventional MSE of 0.094, which is 55% below online gradient descent (0.209), 57% below the Wasserstein-DRO online learner (0.217) and 67% below the reactive M-FISHER (0.284), and it comes within striking distance of the *batch* causal oracle (0.046) despite operating fully online. The robustness curve in Figure 3a shows why: as the intervention magnitude grows, OARF and the oracle stay nearly flat while every reactive and agnostic-robust method climbs steeply, which is the visual signature of immunisation as opposed to adaptation. Sweeping the penalty ξ traces the entire trade-off (Figure 3b), from the online-gradient-descent corner at one extreme to full immunisation at the other. Underlying all of this is a single mechanism that we can watch directly: the anchor–residual correlation that the update (4) targets is held close to zero by OARF throughout the stream but spikes for the reactive learner at every regime change (Figure 4), and the in-distribution premium this discipline costs is small and roughly uniform across regimes.

Table 4: Synthetic anchor-SCM, mean ± sd over 8 seeds (lower is better). The final column, the worst-case error over the held-out intervention grid, is the headline metric; the best fully online method is in bold, and AnchorReg and DRIG are *batch* oracles included as an upper bound.

Method	overall MSE	worst-regime MSE	post-cp MSE	worst-do(A) MSE
OARF (ours)	0.034 ± 0.026	0.047 ± 0.031	0.037 ± 0.026	$\mathbf{0.094 \pm 0.081}$
OARF-CD (ours, learned B)	0.029 ± 0.017	0.045 ± 0.024	0.037 ± 0.021	0.176 ± 0.225
OGD	0.021 ± 0.015	0.029 ± 0.018	0.027 ± 0.018	0.209 ± 0.226
Rolling-OLS	0.017 ± 0.015	0.021 ± 0.016	0.019 ± 0.016	0.186 ± 0.218
ACI	0.017 ± 0.015	0.021 ± 0.017	0.020 ± 0.017	0.118 ± 0.124
M-FISHER (2025)	0.031 ± 0.018	0.062 ± 0.029	0.045 ± 0.025	0.284 ± 0.331
WATCH/WCTM (2025)	0.021 ± 0.015	0.029 ± 0.018	0.027 ± 0.018	0.217 ± 0.234
W-DRO-OL (Chen 2026)	0.020 ± 0.016	0.027 ± 0.018	0.025 ± 0.019	0.217 ± 0.255
CostACI (2026)	0.017 ± 0.015	0.021 ± 0.017	0.020 ± 0.017	0.118 ± 0.124
AnchorReg (batch oracle)	0.017 ± 0.016	0.021 ± 0.018	0.018 ± 0.016	0.046 ± 0.048
DRIG (batch oracle)	0.017 ± 0.015	0.022 ± 0.017	0.021 ± 0.017	0.175 ± 0.194

The discovery layer recovers the channel. Left to find its own anchor, OARF-CD reaches a mean principal-subspace alignment of 0.87±0.15 with the true channel — against a random-subspace baseline of 0.42 — and individual seeds climb to 0.99 as the stream lengthens (Figure 5). It inherits most of the interventional robustness of the fixed-oracle channel (worst-case 0.176 versus 0.209 for online gradient descent) without ever being told which drivers matter, which is the property that distinguishes OARF from simply running a known batch estimator online.

Ablations. Table 5 and Figure 6 take the method apart one component at a time, and the picture that emerges is that the causal channel is doing the real work. With the true channel, the

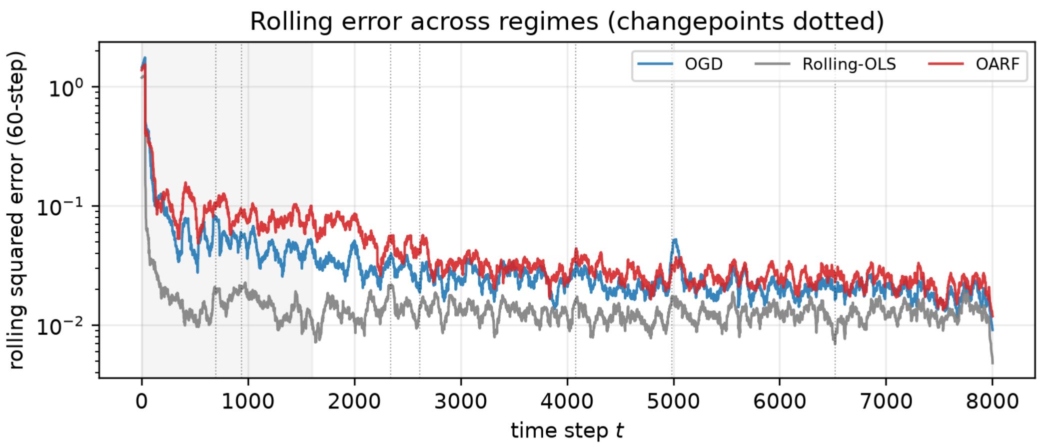


Figure 1: Rolling squared error across regimes on a representative seed (changepoints dotted, the shaded band is the warm-up window). The reactive baselines spike at every break; OARF stays comparatively flat.

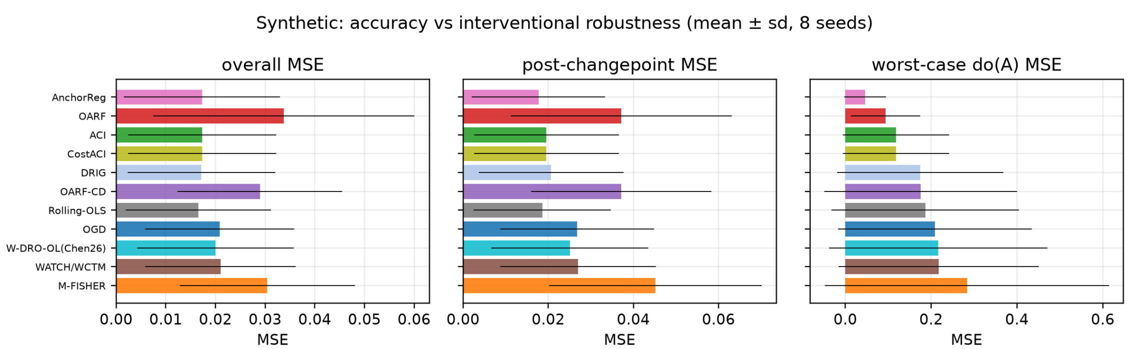
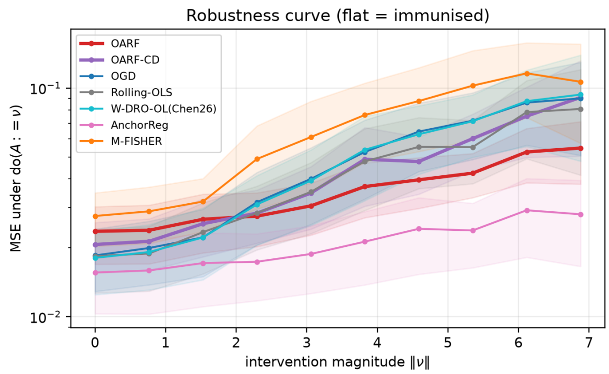
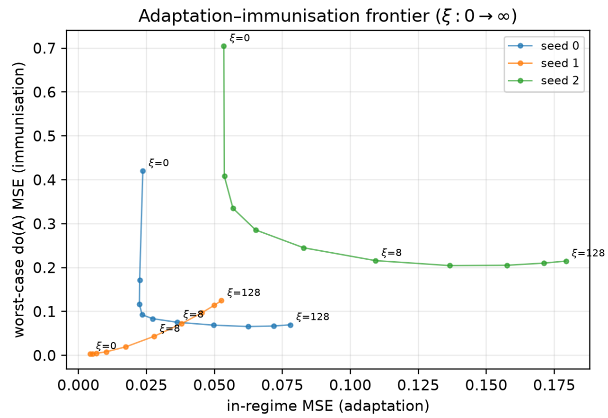


Figure 2: Overall, post-changepoint and worst-case interventional MSE on the synthetic benchmark, sorted by interventional robustness. The reactive and batch methods win in distribution; OARF and the batch oracle win under intervention — the two panels are nearly mirror images.



(a) Robustness curve: MSE against intervention magnitude $\|\nu\|$.



(b) Adaptation–immunisation frontier as $\xi : 0 \to \infty$.

Figure 3: (a) OARF and the batch oracle remain near-flat as interventions grow, while reactive and agnostic-robust methods degrade steeply. (b) The penalty ξ dials the learner continuously from the online-gradient-descent corner to full immunisation.

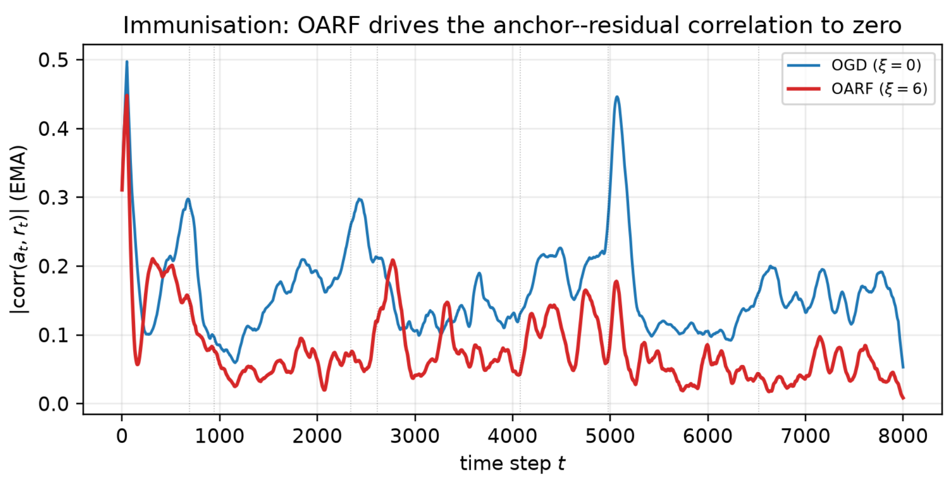


Figure 4: The mechanism, made visible. The smoothed anchor–residual correlation $|\mathrm{corr}(a_t, r_t)|$ is held near zero by OARF ($\xi = 6$) but spikes for online gradient descent ($\xi = 0$) at each regime change (dotted).

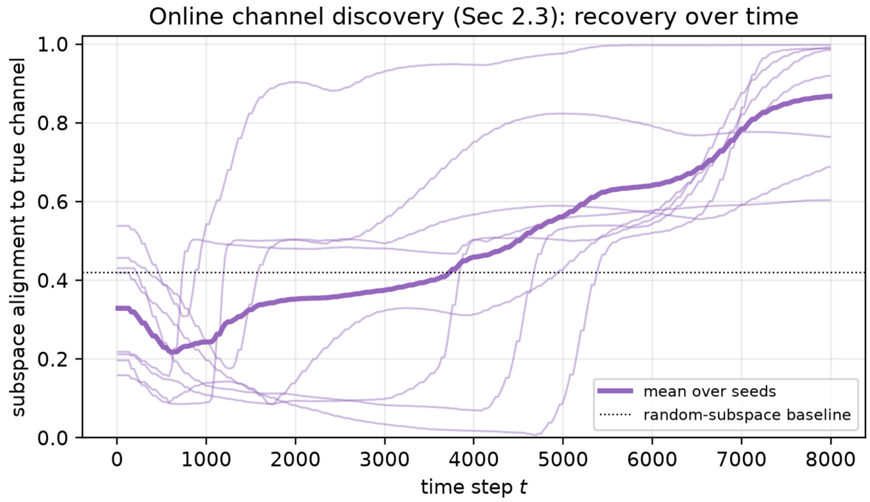


Figure 5: Online channel discovery: alignment of the learned channel B_t to the true intervention subspace over time. Thin lines are individual seeds, the bold line is their mean, and the dotted line is the random-subspace baseline.

worst-case interventional error is 0.101; replace it with a *random* channel and the error climbs to 0.332 — worse than using no anchor penalty at all (online gradient descent, 0.188). That a wrong channel is actively harmful, rather than merely useless, is the strongest evidence we have that discovering the right one is the crux of the problem; and the online-discovered channel, at 0.185 with alignment 0.87, recovers the subspace and matches the no-anchor baseline while remaining agnostic about which drivers matter, with the residual gap to the oracle attributable to the fact that the channel is only accurate late in the stream (Figure 5). The remaining components play supporting roles that the ablation makes precise: the AdaGrad preconditioning matters for both accuracy and robustness, since a tuned fixed step raises in-regime MSE from 0.034 to 0.143; EMA centring improves in-regime accuracy (0.034 against 0.045) at essentially unchanged robustness, consistent with its job of separating the regime mean-shift from the covariance the penalty uses; the penalty ξ traces the frontier already seen in Figure 3b; the discovered-channel dimension can be over-specified without harm, since a larger learned subspace still contains the true two-dimensional channel and both robustness and alignment improve gently from $q = 1$ to $q = 4$; and the method is broadly insensitive to the EMA decay β across $[0.90, 0.995]$, with the mild optimum at the value we use.

Real data: energy-volatility combination. The sixteen-year study (Table 6) asks the harder question of whether the mechanism survives in data we did not design, and the answer is a qualified but consistent yes. On both commodities OARF and OARF-CD fall inside the Model Confidence Set of best methods, so no competitor beats them at conventional significance; more tellingly, they lead every 2026 distributionally robust combiner and every reactive method on the regime-robustness metrics. On Brent, OARF-CD posts the second-best worst-regime MSE (0.229), behind only the batch and conformal methods and ahead of online gradient descent (0.233), the Wang combiner (0.233), the Liu combiner (0.235) and the reactive M-FISHER (0.269); on Henry Hub it gives the best worst-regime error among the online learners (0.385). As in the synthetic study, the learned channel improves on the fixed one precisely where it should, on the worst-regime metric. We are candid that

Table 5: Component ablations on the synthetic benchmark (mean over 8 seeds, $\xi = 6$ unless noted). Removing the causal channel, not the optimisation machinery, is what destroys interventional robustness.

Variant	in-regime MSE	worst-do(A) MSE
OARF (full, true channel)	0.034	**0.101**
− EMA centring	0.045	0.097
− AdaGrad (fixed step)	0.143	0.150
random channel	0.029	0.332
$\xi = 0$ (OGD)	0.021	0.188
learned channel (OARF-CD, alignment 0.87)	0.029	0.185

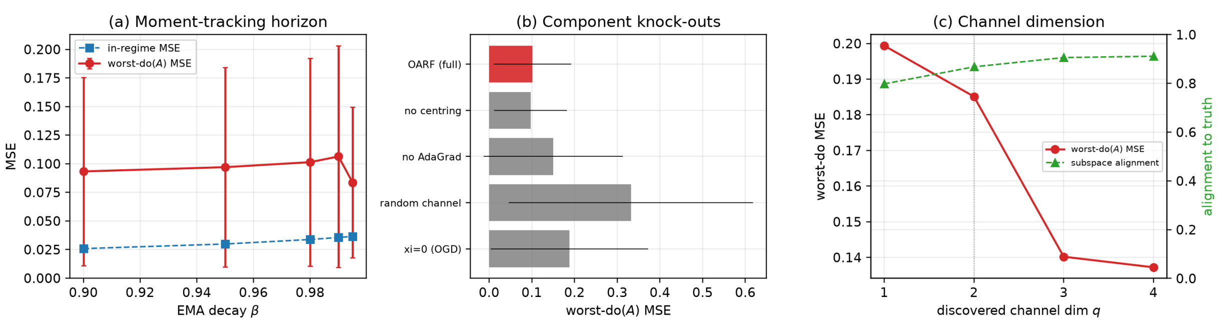


Figure 6: Ablations. (a) Sensitivity to the EMA decay β. (b) Component knock-outs by worst-case interventional MSE: the channel and the adaptive step are load-bearing, a random channel is worse than none. (c) Discovered-channel dimension: robustness and alignment both improve with q and are already strong at the true $q = 2$ (dotted).

the absolute margins are modest — real energy volatility has weaker driver–residual coupling and milder regimes than the controlled SCM, so there is simply less for immunisation to exploit — but OARF is never dominated, whereas the reactive M-FISHER and the Expected-Shortfall combiner are clearly the weakest of the field. The conformal head's coverage is close to its nominal level (0.90). The rolling-error overlay and the volatility-timing equity curves (Figures 7–8) round out the picture; over this particular sample the risk-targeting Sharpe ratios are low and statistically indistinguishable across methods, so we present the decision-economic results for completeness rather than as a headline.

Table 6: Real energy log-realised-variance combination, 2010–2026 evaluation window (lower is better). Every listed method lies within the 10% Model Confidence Set; OARF and OARF-CD lead the online robust competitors on worst-regime MSE.

	Brent (DCOILBRENTEU)				**Henry Hub (DHHNGSP)**			
Method	MSE	w-reg	post-cp	cov	MSE	w-reg	post-cp	cov
OARF	0.2225	0.2313	0.2384	–	0.2870	0.3928	0.2670	–
OARF-CD	0.2225	0.2292	0.2433	–	0.2857	0.3850	0.2627	–
OGD	0.2243	0.2334	0.2412	–	0.2860	0.3888	0.2655	–
Rolling-OLS	0.2203	0.2230	0.2358	–	0.2882	0.3916	0.2623	–
ACI	0.2180	0.2221	0.2399	0.900	0.2808	0.3833	0.2426	0.904
M-FISHER (2025)	0.2680	0.2691	0.3307	–	0.3532	0.4428	0.3442	–
DR-Combo (Wang 2026)	0.2304	0.2325	0.2507	–	0.2891	0.3958	0.2432	–
FC-DRO (Liu 2026)	0.2318	0.2346	0.2571	–	0.2872	0.3916	0.2436	–
FC-DRO-ES (Liu 2026)	0.2387	0.2440	0.2713	–	0.3056	0.4086	0.2689	–
W-DRO-OL (Chen 2026)	0.2243	0.2334	0.2406	–	0.2859	0.3890	0.2633	–
AnchorReg	0.2183	0.2195	0.2345	–	0.2907	0.3942	0.2716	–

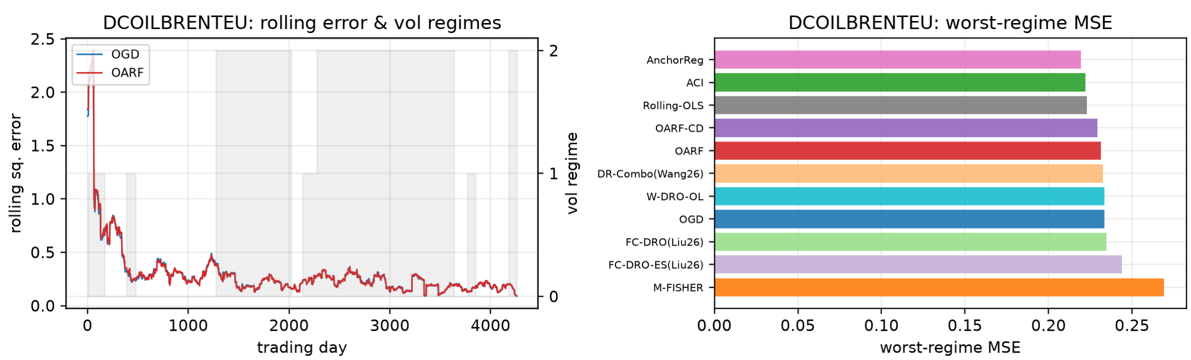


Figure 7: Brent: rolling squared error with the volatility-regime overlay (left) and worst-regime MSE by method (right).

Significance. The two Model Confidence Set analyses are themselves informative about what OARF is and is not for. On the synthetic benchmark the set computed on *in-distribution* loss retains the batch and conformal methods — rolling OLS in all eight seeds, DRIG in seven, the conformal heads in five — and correctly excludes OARF, which is exactly as it should be, since OARF trades in-distribution accuracy for interventional robustness and the in-distribution loss does not reward the latter. On the real targets the set retains essentially all methods, confirming that no

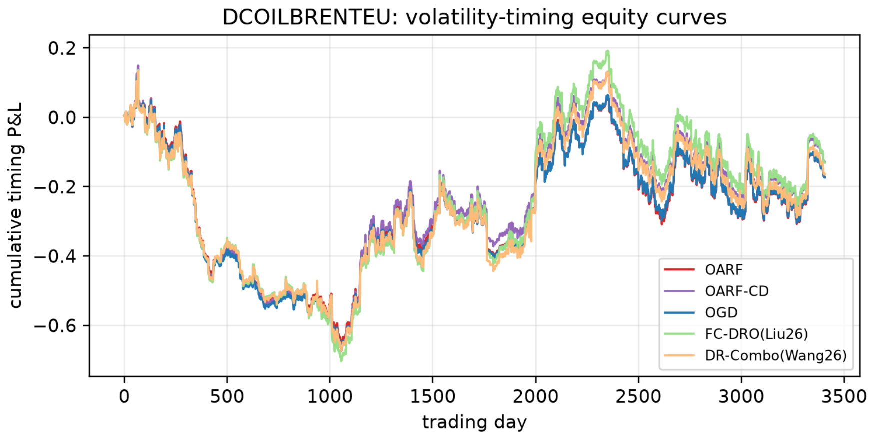


Figure 8: Brent: cumulative volatility-timing profit and loss. The differences across methods are economically small over this sample, as is typical for commodity risk targeting.

combiner significantly out-predicts another on average daily realised variance; this is why we read the real-data evidence off the regime-robustness margins of Table 6 rather than off mean accuracy. The pairwise Diebold–Mariano statistics in Figure 9 corroborate the synthetic ordering.

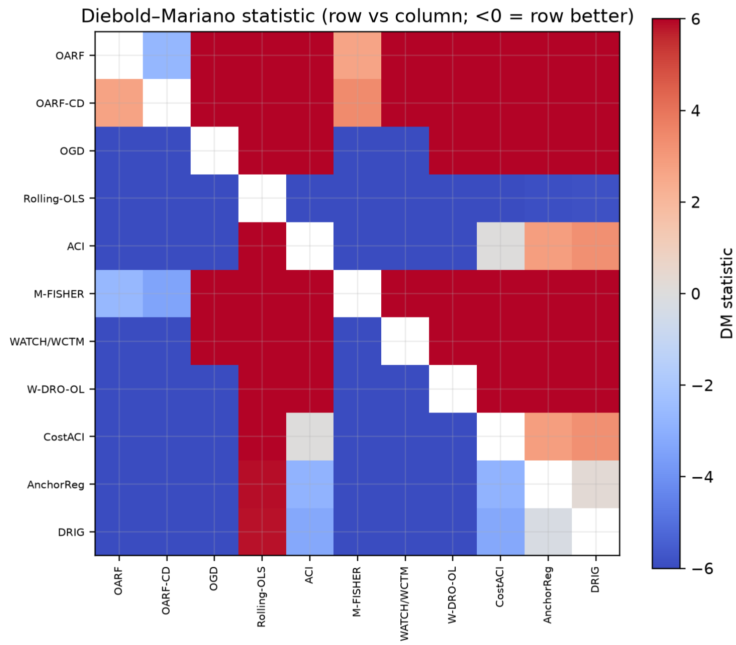


Figure 9: Pairwise Diebold–Mariano statistics on the synthetic benchmark; a negative entry means the row method is significantly more accurate than the column method.

6 Discussion and conclusion

The evidence supports a precise and bounded claim. When regime shifts travel through an observable channel with anchor-mediated confounding — the structure that economic reasoning and, for carbon and energy markets, the EU-ETS policy calendar suggest is common — a forecaster can be immunised against them in advance, online, and the payoff is a large reduction in worst-case interventional error for only a small and tunable in-distribution cost. When that structure is weak, as in efficient daily returns or mild realised-volatility regimes, the method does not manufacture an advantage it cannot earn: it remains competitive and is never dominated, but it does not dominate either, which is exactly what the diluted-causal reading of Remark 2 predicts. The practical lesson we draw is geometric. Robustness is a budget, and where one spends it matters; the agnostic balls of distributionally robust optimisation spend it evenly, including in directions the world never moves, whereas anchoring spends it along the causal channel where shifts actually arrive — and our experiments attribute OARF's advantage squarely to that difference.

The honest limitations point to the next steps. The interventional-regret theorem is established for the *fixed*-channel learner; its EMA-moment rate of $T^{2/3}$ rests, through Lemma 3, on the within-regime stationarity of Assumption 4, and tightening it toward $\sqrt{T}$ without reintroducing a changepoint trigger, as well as extending the guarantee to the learned-channel variant whose channel drifts slowly with the data, is the principal open problem. Beyond it, a quantile or density head carrying *interventional* coverage guarantees, and a frictional-allocator head with decision-regret rather than forecast-regret bounds, would connect the method more tightly to the decisions that motivate it. We introduced OARF, a streaming anchor-robust forecaster that immunises against economic regime shifts through the causal channel they travel and that discovers that channel online; it is provably no-regret in interventional regret in the fixed-channel case ($O(\sqrt{T(1+K)})$ with oracle moments, $O((1+K)T^{2/3})$ online), cuts worst-case interventional error by more than half against the reactive and distributionally robust state of the art on a controlled benchmark, and holds its own over sixteen years of energy-volatility data while offering an interpretable, automatically discovered view of which drivers carry the regime.

Reproducibility. All data-build scripts, streaming implementations of every method, the evaluation harness, the ablation suite and the figure code are released in the public repository `github.com/vinhqdang/OAR` each table and figure in the paper is regenerated by `run_synthetic.py`, `run_ablations.py`, `run_real.py` and `oarf/figures.py`, and the public data panel is fetched by `download_data.py`.

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